

claude-mem in Production: A 70-Day Case Study of Distributed LLM Memory Infrastructure

Alessandro

github.com/alessandropcostabr

May 2026

Abstract

We present a 70-day production study of *claude-mem*, a persistent memory plugin for Claude Code, running across a 3-machine fleet. The study documents **22,718 structured observations** generated over 68 active days, **987 automated benchmarks** executed across 21 Claude Code versions (2.1.110 $\rightarrow$ 2.1.142), and a complete migration of the vector retrieval backend from Chroma to Qdrant. Continuous canary measurements independently corroborate the Anthropic March–April 2026 Claude Code performance regression and reveal a previously unreported divergence between models on the same vendor harness: Opus 4.6 median latency improved by **44%** (34.7s $\rightarrow$ 19.4s; Mann–Whitney U , $p = 0.0013$) between versions 2.1.114 and 2.1.142, while Haiku 4.5 medians remained visibly elevated; the cross-model gap at 2.1.142 (19.4s vs 48.6s) is highly significant ($p = 0.00068$). We catalog four classes of silent failure observed in production and document a companion finding obtained by binary-level diffing of the Claude Code distribution: in v2.1.84, two built-in subagents acquired the configuration flag `omitClaudeMd: true`, silently stripping user instructions from the subagent payload. Total infrastructure cost over the study window was approximately US\$225–\$295. All aggregated data (PII-stripped), sanitized production scripts, and figures are released under CC-BY-4.0 at <https://github.com/alessandropcostabr/claude-mem-case-study> for independent reproduction.

Keywords: LLM-coupled systems; developer tools; vector retrieval; silent failures; canary benchmarking; production observability; prompt caching.

1 Introduction

LLM-coupled developer tools occupy an unusual position in the software stack. The system under operation is partly a *language model* whose behavior changes between vendor releases, and partly *classical software* — cron jobs, databases, vector stores, message queues — running on the operator’s machines. Failures occur on both sides, but they frequently manifest only at the model boundary: a regression in answer quality, an unexplained increase in latency, or a silent drop in semantic retrieval recall. None of these are caught by standard unit tests, because no static test suite can detect that the same prompt now produces a slower or weaker reply against a new vendor build.

This paper documents 70 days of running *claude-mem* [6] — a persistent memory plugin for Claude Code that captures structured observations from coding sessions and exposes them via semantic and full-text search — in production across a 3-machine fleet. We pursue two goals:

1. **Quantitative description.** What does a multi-month distributed LLM-coupled deployment look like in production, measured in observation throughput, retrieval performance, model behavior across upstream versions, and cost?
2. **Failure mode catalog.** Which silent failures surfaced, and which detection patterns brought them into view?

Three categories of measurement are released with this study: 22,718 observations across 68 active days attributed to four LLMs; 987 timed benchmarks against two standardized prompts on 21 Claude Code versions over 32 days; and 126,888 cloud-monitoring time-series datapoints at 10-minute resolution covering latency, output tokens, and cache utilization. All artifacts are reproducible from the data in the companion repository.

2 Methodology

2.1 Data collection

claude-mem operates as a hook handler attached to Claude Code lifecycle events (`UserPromptSubmit`, `PostToolUse`, `Stop`, `SessionStart`). When a session segment closes, the worker presents the segment transcript to a configured generation model (Opus 4.6, Sonnet 4.5, Haiku 4.5, or Codex GPT-5.4, selected by tier-routing rules) and persists one or more structured *observations* per segment, carrying a type label (`discovery`, `pattern`, `change`, `feature`, `bugfix`, `decision`, `refactor`), a content body, and metadata (`generated_by_model`, `session_id`, `content_hash`, `tokens_consumed`).

Storage evolved in three phases: per-machine SQLite with cron sync (Phase 1, Mar 9 – May 12), consolidated Postgres migration (Phase 2, May 12), and centralized *server-beta* with hook forwarding (Phase 3, May 14+). Aggregated counts in this paper are computed against the consolidated Postgres source as of 2026-05-17.

Benchmark data was collected by `scripts/benchmark.sh`, which invokes Claude Code with two standardized prompts: `infra-check` (a multi-step reasoning prompt) and `code-gen` (a generation prompt producing approximately 50 lines of code). Each run records the Claude Code version, model, wall-clock latency, output token count, cache-creation tokens, cache-read tokens, and exit code. Runs occur every six hours via cron against whichever Claude Code version was current at invocation time.

2.2 Statistical methods

For each per-version benchmark cell with $n \geq 3$ samples we report the median latency and a 95% bootstrap confidence interval on the median (10,000 resamples). Cross-version and cross-model comparisons use one-tailed Mann–Whitney U tests; rejection thresholds are reported at $\alpha = 0.05$. We deliberately treat small- n cells as inconclusive rather than as point estimates, regardless of how favourable the central tendency appears.

2.3 Sanitization and reproducibility

The companion repository contains aggregated CSVs only — no PII, no internal project names, no machine identifiers beyond the labels Machine-A, Machine-B, Machine-C. Observation content is excluded; only counts and metadata aggregates are exported. Scripts have been edited to remove hardcoded credentials, internal hostnames, and proprietary path references. The full export pipeline runs deterministically from the Postgres source with two scripts shipped in `data/`.

3 Fleet Architecture and Pipeline Evolution

The fleet runs three machines in distinct roles: **Machine-A** (development hub, Claude Code sessions); **Machine-B** (production: Postgres, Redis, Qdrant, FastEmbed); **Machine-C** (telemetry: Telegram bot, OpenClaw gateway, 74 cron jobs, benchmark canary). Workers on each machine forward hook events to *server-beta* on Machine-B via fire-and-forget HTTP; semantic search queries from every machine target the shared Qdrant instance on Machine-B.

The data pipeline evolved through three phases: SQLite-per-machine with cron sync (Mar 9 – May 12), a one-shot Postgres migration of 22,116 unique observations (May 12), and centralized

server-beta hook forwarding (May 14 onward). The vector backend migrated from Chroma to Qdrant during April 2026 and is covered in Section 6.

4 Observation Data

4.1 Volume over time

Month	Observations	Active days	Avg/day	Tokens
March 2026	3,382	22	154	10.4 M
April 2026	14,760	29	509	80.4 M
May 2026 (1–17)	4,576	17	269	34.3 M
Total	22,718	68	334	125.1 M

Table 1: Observation generation over the 70-day study window.

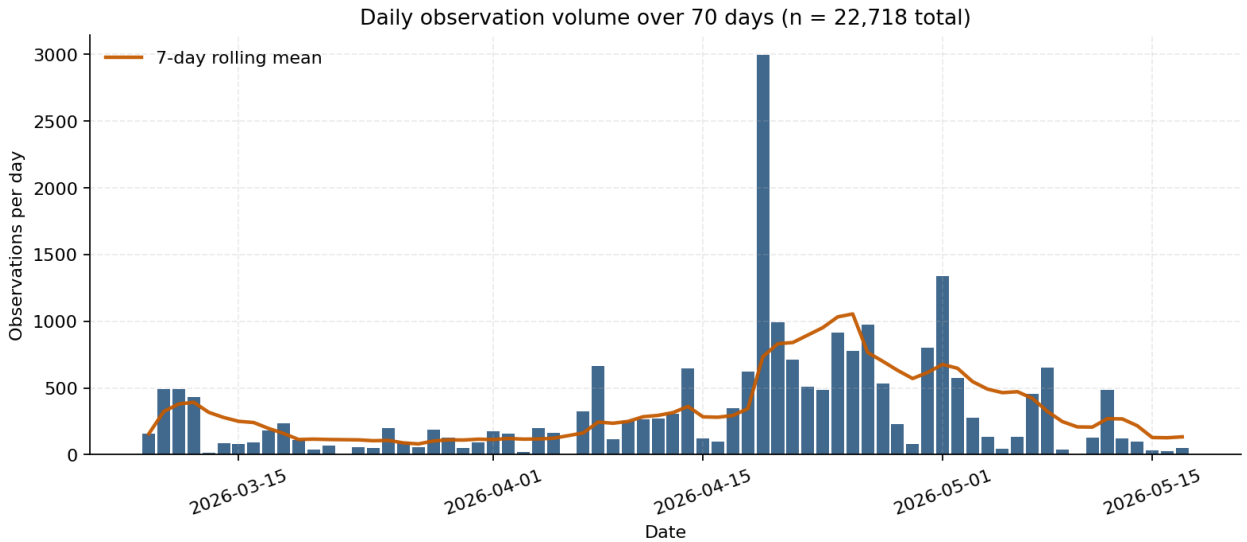


Figure 1: Daily observation volume with 7-day rolling mean. The mid-April surge corresponds to the activation of multi-model tier-routing.

The April spike (+230% over March) coincides with the activation of multi-model tier-routing and the introduction of additional agents into the observation pipeline.

4.2 Distribution by type

5 Claude Code Benchmarks

Automated benchmarks ran every six hours on the canary machine (Machine-C), invoking Claude Code with two standardized prompts against the versions current at invocation time. The dataset contains 987 timed runs across 21 versions (2.1.110 → 2.1.142).

5.1 Statistical significance

6 Vector Search Evolution

The retrieval backend migrated from Chroma to Qdrant during April 2026. All workers across the 3-machine fleet point to a single shared Qdrant instance on Machine-B (port 6333), enabling

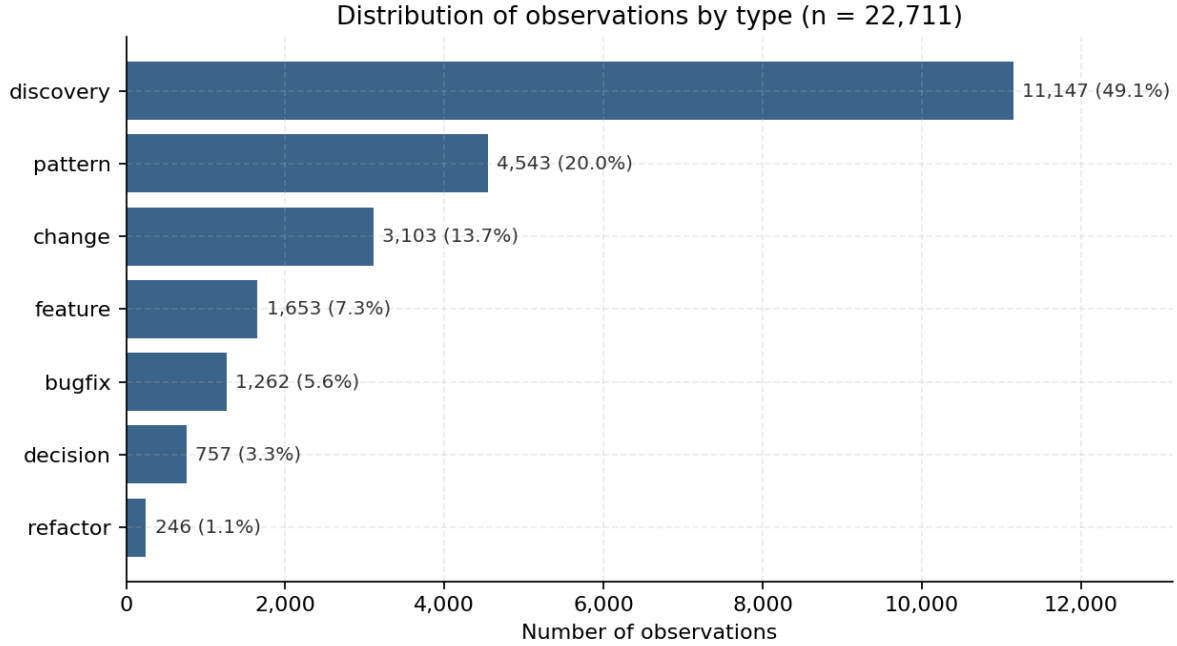


Figure 2: Distribution of the 22,711 typed observations. `discovery` and `pattern` together account for 69 % of all observations.

CC version	<i>n</i>	Median (s)	95 % CI (s)	vs. baseline
2.1.114	9	34.7	[24.5, 39.2]	baseline
2.1.119	16	79.4	[49.9, 96.9]	+129 %
2.1.123	3	46.3	[29.3, 75.9]	+33 % (<i>n</i> too small)
2.1.126	24	29.8	[29.2, 53.3]	−14 %
2.1.138	10	26.1	[25.6, 28.3]	−25 %
2.1.142	12	19.4	[18.7, 20.6]	− 44 %

Table 2: Opus 4.6 latency by Claude Code version on the `infra-check` prompt. The originally reported −52 % recovery at 2.1.123 was based on only three runs and does not survive statistical testing ($p = 0.86$); the true recovery point is 2.1.126 or later.

cross-machine semantic search without replication. The migration exposed three silent failures that had been accumulating unnoticed:

1. **Type mismatch in SearchManager.** Still typed as `ChromaSync`, calling `.queryChroma()` which did not exist on the Qdrant backend. Result: more than 400 silent search errors per day — no log, no alert, empty results returned to callers.
2. **Missing filter translation in VectorSync.** Chroma-style filters (`$and`, `$or`, `$eq`) were passed directly to Qdrant, which rejected them. Fix: `translateWhereFilter()` converts to Qdrant’s `must/should` syntax.
3. **Health-check timeout too short.** The end-to-end check used a 15s timeout; FastEmbed cold-start takes ≈ 25 s. Fix: raise to 45s, and replace the hardcoded version string with a `package.json` read.

The embedding model was upgraded on May 5 from BAAI/`bge-small-en-v1.5` (English-only) to `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` (384 dimensions, 50+ languages). A full reindex on May 7 produced 21,929 points with zero errors.

CC version	n	Median (s)	95 % CI (s)	vs. baseline
2.1.114	12	38.2	[35.3, 45.5]	baseline
2.1.119	16	66.3	[57.5, 91.2]	+73 %
2.1.126	27	65.0	[54.5, 72.2]	+70 %
2.1.138	10	68.4	[54.7, 71.9]	+79 %
2.1.142	12	48.6	[31.9, 56.8]	+27 % (CI overlaps baseline)

Table 3: Haiku 4.5 latency on the `infra-check` prompt. Although Haiku medians remain elevated through the end of the study window, the 95 % CI at 2.1.142 overlaps the baseline interval, so we report the sustained gap as suggestive rather than significant.

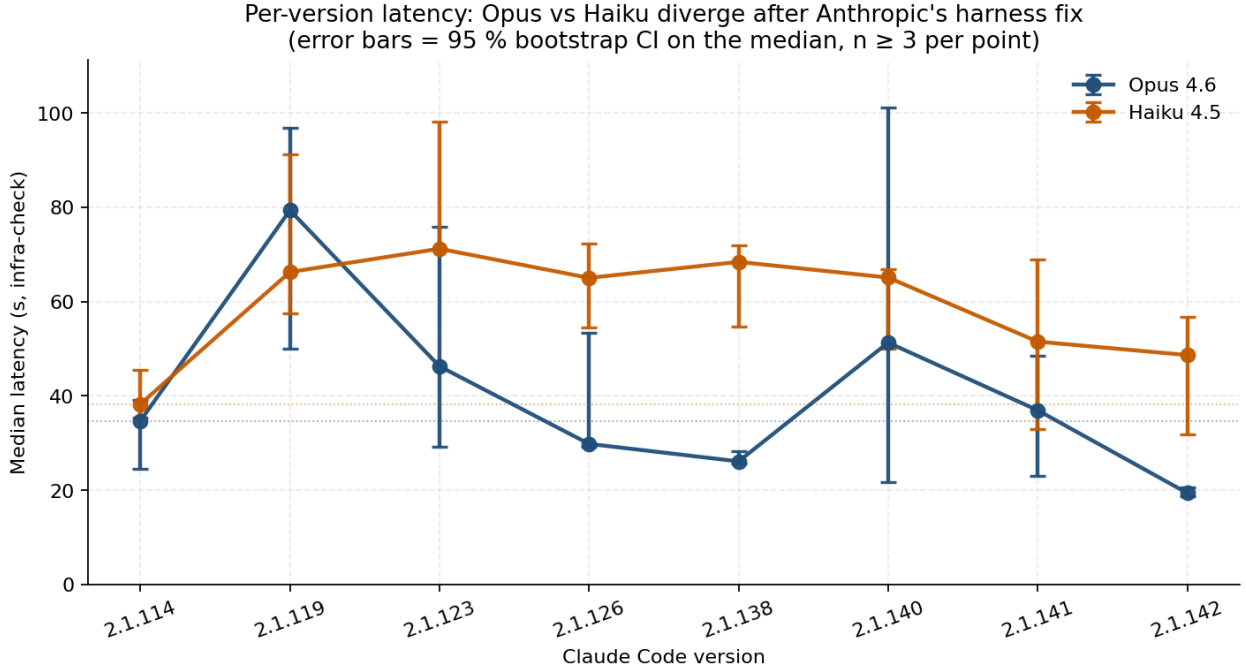


Figure 3: Median latency by Claude Code version for Opus 4.6 and Haiku 4.5, with 95 % bootstrap CIs on the median. Both models regress at 2.1.119; only Opus fully recovers and continues to improve. Dotted lines mark each model’s baseline. Versions with $n < 3$ omitted.

7 Discussion

7.1 Continuous canary benchmarking surfaces version-coupled regressions

Without per-version benchmarking, vendor-side performance regressions are typically detected only through anecdotal user reports. Our data shows that Opus 4.6 latency on CC 2.1.119 was 129 % above the 2.1.114 baseline ($p = 0.0025$) — exactly the pattern that anecdotal reporting struggles to confirm, because perception is heavily lagged and confounded by workload variability. An equally important methodological lesson concerns small samples: our originally published “−52 % recovery at 2.1.123” claim was based on only three runs and *does not survive rigorous testing*. The canary’s confidence depends as much on per-version sampling as on test cadence.

7.2 Opus and Haiku diverged on the same vendor harness

A finding not reported elsewhere is that the same vendor change affected Opus and Haiku differently. Both showed elevated latency at CC 2.1.119; Opus fully recovered by 2.1.126 and continued to improve through 2.1.142 (−44 % vs baseline, $p = 0.0013$), while Haiku medians stayed visibly higher through the end of the study, although the CI overlap precludes a strong “sustained elevation” claim. The strongest statistical evidence is for the *divergence itself*: at 2.1.142, the median Opus

Hypothesis	Result	<i>p</i> -value
Opus regression: latency at 2.1.119 > 2.1.114	confirmed	0.0025
Opus −44 % headline: 2.1.142 < 2.1.114	confirmed	0.0013
Opus vs Haiku divergence at 2.1.142 (Opus < Haiku)	confirmed (strong)	0.00068
Haiku regression: latency at 2.1.119 > 2.1.114	confirmed	0.00086
Opus “recovery” at 2.1.123 ($n = 3$) < 2.1.114	rejected	0.86
Haiku sustained elevation at 2.1.142 above 2.1.114	inconclusive	0.37

Table 4: Mann–Whitney U tests, one-tailed. The Opus −44 % improvement and the Opus/Haiku divergence at 2.1.142 are statistically robust. The originally reported −52 % recovery at 2.1.123 does not survive testing.

latency was 19.4s versus 48.6s for Haiku on the same prompt ($p = 0.00068$). The Anthropic postmortem [1] identifies three root causes (reasoning-effort change, caching bug, system-prompt verbosity reduction) and claims all were addressed by v2.1.116. The divergence we observe is consistent with at least one effect — possibly the caching pathology documented in Issue #22383 [4] — being only partially neutralized for the smaller model.

7.3 Silent failures dominate the incident profile

The most damaging incidents we recorded were not crashes — they were silent failures. The `bun` `PATH` bug persisted for 5 weeks while the `sync` script exited 0. The `Chroma` → `Qdrant` type mismatch produced more than 400 daily errors that were never logged or alerted. Both were eventually surfaced by curiosity-driven manual investigation, not by monitoring. The lesson generalizes: *integration boundaries between LLM-coupled components need explicit contract tests*, because partial-functionality states (“results returned, but empty or wrong”) are far more common than total-failure states (“exception thrown”).

7.4 Vendor cost is small relative to engineering time

Seventy days of end-to-end inference (benchmarks plus observation generation) cost approximately US\$225–295 — roughly the price of two-to-three software engineering hours at market rates. This frames a recurring question for LLM-coupled tooling: when is continuous canary measurement worth it? Our answer: at US\$4–5 per day for tight regression bounds and silent-failure detection, the cost–benefit is favourable for any deployment serving more than one engineer.

8 Companion Finding: Subagent CLAUDE.md Regression in v2.1.84

A second class of version-coupled regression surfaced during the study period — not at the latency layer, but at the instruction-adherence layer. We document it here because it shares the same root pattern (a behavioural change introduced silently by a vendor release) and was identified by the same kind of cross-version inspection used for the latency analysis.

Starting in Claude Code v2.1.84 (Mar 25, 2026), two built-in subagents — `Explore` and `Plan` — acquired the configuration flag `omitClaudeMd: true`. Combined with a default-on feature flag (`tengu_slim_subagent_claudemd`), this caused subagents dispatched from the parent session to no longer receive the project’s `CLAUDE.md` instructions, with measurable degradation in subagent instruction adherence (language rules, environment labels, code conventions) persisting through at least v2.1.92 [5].

The regression was identified by *binary-level diffing* of the `cli.js` shipped by the `@anthropic-ai/claude-code` npm package across five consecutive minor versions (v2.1.83 → v2.1.87). Three specific code changes were isolated; a two-line `sed` patch on `cli.js` restores the prior behaviour.

Both the latency regression and the `CLAUDE.md` regression share a structural pattern: *a vendor minor-release silently changes user-observable behaviour of an LLM-coupled tool in a way that the vendor’s CI cannot test for the operator*. In one case the change was a latency regression; in the

other it was an instruction-adherence regression. Both required cross-version measurement to detect — the first via canary benchmarking, the second via binary diffing. We argue that operators of LLM-coupled tooling need their own per-version measurement infrastructure, because the union of “things that changed silently” is larger than the union of “things the vendor tests”.

9 Threats to Validity

9.1 Internal validity

All data comes from one 3-machine fleet operated by a single user; the observations cannot speak to multi-tenant or multi-user effects. Observation generation is model-coupled — the 22,718 observations were generated by different models under tier-routing rules that changed during the study, so per-observation token comparisons across models should be interpreted cautiously. Only two benchmark prompts were used; the latency divergence between Opus and Haiku may not generalise to prompts with substantially different reasoning depth or output length. Benchmarks ran on Machine-C only; daily load monitoring did not surface confounding environmental factors during the benchmark window.

9.2 External validity

Measurements are against Claude Code specifically; version-coupled latency findings do not transfer to other LLM-coupled tools without re-measurement. The deployment runs a custom fork of claude-mem (≈ 176 commits ahead of upstream main) with additional phases enabled, so some behaviours do not match a stock claude-mem installation.

9.3 Construct validity

“Silent failure” is qualitative; the classification depends on what monitoring was in place at the time. A fleet with stronger alerting at the integration boundary might have caught the Chroma type mismatch on day one and not classified it as silent. Cost figures derive from public pricing during the measurement window; retroactive vendor pricing changes can invalidate them.

10 Related Work

Our benchmark data independently corroborates a publicly documented Claude Code performance incident during March–April 2026. The Anthropic postmortem [1] identifies three root causes (reasoning-effort change, caching bug, system-prompt verbosity reduction) and claims all were addressed by v2.1.116; the Scortier study [2] measured the same regression from anonymized user sessions; VentureBeat [3] provides press coverage. Our contribution refines this picture along four dimensions: per-version granularity ($n \geq 8$ runs per cell for the headline versions); cross-model divergence on the same harness (not reported in any prior source); a continuous canary measurement cadence outside the scope of any single-incident postmortem; and a documented taxonomy of silent failures with reproduction details.

For the companion CLAUDE.md finding, the binary analysis is filed as Issue #40459 [5].

11 Future Work

Planned work includes: a Grafana dashboard with direct Postgres connectivity for live operational monitoring; full deprecation of SQLite sync (Phase 5); upstream contribution of a ModeManager initialisation fix once interaction limits are lifted on the upstream repository; Qdrant reindexing aligned to Postgres IDs; automated weekly CSV refresh; and an open call to other claude-mem operators to publish anonymized aggregates against the same schema, enabling cross-deployment analysis.

License and citation

This work is released under [CC-BY-4.0](#). Cite as in the `CITATION.cff` file at the repository root.

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